Multiview Spatial–Spectral Two-Stream Network for Hyperspectral Image Unmixing

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***Abstract*— Linear spectral unmixing is an important technique in the analysis of mixed pixels in hyperspectral images. In recent years, deep learning-based methods have been garnering increasing attention in hyperspectral unmixing; especially, unsupervised autoencoder (AE) networks that have achieved excellent unmixing performance are a recent trend. While most approaches use spatial information, it is well known that hyperspectral data are characterized by a large number of narrow spectral bands. In order to take full advantage of the hyperspectral bands in unmixing and the spatial information, in this article, we explore multiview spectral and spatial information in an AE-based unmixing framework. We introduce multiview spectral information through spectral partitioning and propose a multiview spatial–spectral two-stream network, called MSSS-Net, which simultaneously learns a spatial stream network and a multiview spectral stream network in an end-to-end fashion for more efficient unmixing. The MSSS-Net is a two-stream deep unmixing network sharing a decoder, where its two AE networks employ recurrent neural networks (RNNs) to collaboratively utilize multiview spectral and spatial information. The spatial stream network branch extracts the spatial features of pixels and its neighbors, while the multiview spectral stream network branch exploits the multiview spectral bands of a pixel. Meanwhile, we design a cascaded bidirectional and unidirectional RNNs' encoder structure for multiview spatial–spectral information to learn more discriminative deep patch-pixel features. Extensive ablation studies and experiments on both synthetic and real datasets demonstrate the superiority of the MSSS-Net over state-of-the-art unmixing methods.**

***Index Terms*— Autoencoder (AE), hyperspectral imaging, multiview learning, recurrent neural network (RNN), spatial–spectral unmixing.**

I. Introduction

DUE to the constraints between spectral and spatial resolution of hyperspectral imaging sensors, coupled with complex surface and light scattering, mixed pixels, which are comprised of multiple materials spectra, are ubiquitous in a hyperspectral image (HSI) [1]. Mixed pixels have become challenging in HSI processing and applications, such as classification, detection, and fusion [1], [2]. Therefore, in the utilization of hyperspectral data, endmembers and their abundances should be extracted and estimated from HSI. This procedure is known as hyperspectral unmixing (HU) or mixed pixel decomposition [2]. Based on specific mixing models, unmixing algorithms can be linear or nonlinear [3], [4]. Considering its simplicity and physical interpretability, the linear mixed model (LMM) is the most widely used, and it assumes that an observed spectral pixel is a linear combination of endmembers and their respective abundances [3]. Intuitively, abundances should satisfy two physical constraints, i.e., the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC) [5].

In this work, we concentrate on linear unmixing. Under LMM, traditional unmixing methods are mainly based on geometry, statistics, or sparse regression. The vertex component analysis (VCA) [6], N-FINDR [7], and the iterative error analysis (IEA) [8] are the best-known geometrical-based unmixing algorithms. However, the pure pixel assumption restricts the performance of these algorithms in real HSIs. Minimum volume-based methods, such as the minimum volume simplex analysis (MVSA) [9], are proposed to process highly mixed HSIs. On the other hand, statistical algorithms are alternative methods that efficiently handle highly mixed pixels. Unmixing methods based on the Bayesian framework and nonnegative matrix factorization (NMF) are typical representatives. The former utilizes statistical suppositions and prior knowledge to improve the inference performance of unmixing process [10], while the latter can simultaneously estimate endmembers and abundances [11]. Extended NMFs with regularization constraints have been raised to incorporate spatial and spectral information for improved unmixing accuracy [12], [13], [14], [15].

Sparse unmixing [16], [17] is another important type of unmixing algorithm, which is proposed as a semisupervised approach based on a large preknown spectral library. The proposal of a sparse unmixing algorithm via variable splitting and augmented Lagrangian (SUnSAL) [16] has promoted the prosperity of sparse unmixing methods, which imposes the classical l_1 regularization on the estimated abundance matrix. Thereafter, by employing the total variation (TV) and $l_{2,1}$ regularization terms, the SUnSAL-TV [18] and collaborative

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SUnSAL (CLSUnSAL) [19] algorithms enhance the spatial characteristics of neighborhood pixels and the row sparsity of abundances, respectively. In addition, weighted sparse unmixing algorithms are also widely adopted due to their good unmixing performance. The weighted l_1 norm is proposed in [20], which penalizes the nonzero coefficients in estimated abundance by a weighting factor. Meanwhile, a diversity of spectral and/or spatial weighted regularizers [21], [22], [23], [24] are successively introduced to promote the positive effect of spatial–spectral information on unmixing.

In recent years, deep learning-based approaches have become popular in HU, especially the deep autoencoder (AE) network framework and its variants. The AE model learns a low-dimensional representation through an encoder to obtain abundances and then uses a decoder to reconstruct the HSI, where the reconstruction weights are endmembers. AE can be effectively trained unsupervised with constraints such as ANC and ASC [25]. uDAS [26], DAEN [27], DeepGUn [28], EndNet [29], TANet [30], SSRDMF [31], MNN-BU [32], and SNMF-Net [33] are representative examples built on this framework. uDAS first proposes an untied encoder and decoder, and imposes different constraints on them. The use of denoising AE and $l_{2,1}$ norm constraint enables the AE-based unmixing framework to achieve good results. DAEN constructs stacked AEs to deal with outliers and noise, and employs a variational AE (VAE) to constrain abundances, while, in DeepGUn, VAE is adopted to learn a spectral variability model to cope with the endmember variability in HSI. EndNet attempts to improve the original AE network by introducing additional layers, metrics, and new loss functions to obtain better solutions and sparser estimates. Similarly, TANet improves the AE network and introduces another AE with the same structure to enhance the physical meaning of endmembers. Differently, SSRDMF proposes a deep matrix factorization network based on AE, which adopts a multilayer nonlinear encoder and introduces a self-supervised constraint, while, in MNN-BU and SNMF-Net, the learning-based AE structure and the model-based unmixing method are combined together, and a good unmixing effect is achieved.

In the above AE-based unmixing approaches, the training of networks is pixel-by-pixel, which ignores the spatial information in HSI. Hence, cubewise or patchwise unmixing networks have been proposed successively. In [34], the convolutional neural network (CNN) is introduced into the AE model. It directly utilizes the patches of HSI for unmixing to preserve spatial features. Inspired by the perception mechanism, a cycle-consistency AE model is proposed in [35], which performs efficient unmixing by learning two cascaded AEs. Spatial–spectral unmixing can be performed by changing the convolution filter sizes, while, in [36], a Siamese unmixing model is presented, which corrects the estimated endmembers through another AE network. Similarly, it can also be applied to model spatial information through convolution operations. Most recently, a Bayesian fully convolutional unmixing model is proposed in [37], taking into account factors such as spatial information, noise, and endmembers. In [38], a minimum simplex convolutional network is presented, which introduces convolutional filters and geometrical information. From a different perspective, a spatial–spectral collaborative unmixing network is proposed in [39], which designs a spatial AE and a spectral convolutional AE to collaboratively utilize spatial and spectral information.

Reviewing conventional unmixing approaches and popular AE-based unmixing networks, it can be observed that most methods, including [13], [14], [15], [18], [21], [22], [23], and [24] in traditional methods and [34], [35], [36], [37], [38], and [39] in AE-based methods, mainly focus on exploring spatial features in hyperspectral data. Spatial information is crucial in the unmixing task, and its importance has been confirmed in numerous studies [18], [34], [39]. How to efficiently utilize spatial information, as well as spectral information, is the key to improve the estimation accuracy. Therefore, we propose a spatial–spectral two-stream network to fully collaboratively exploit the spatial–spectral information. Meanwhile, we design a cascaded recurrent neural networks (RNNs) encoder to utilize spatial information more efficiently. On the other hand, while focusing on spatial information, we know that the advantage of HSI is that each mixed pixel has a continuous spectral dimension, which contains bands with 10-nm resolution in the 0.3–2.5-μm spectral range. The wealth of spectral information available in HSI is needed to serve high-quality unmixing.

To exploit the rich spectral information, spectral partitioning techniques have actually been widely used in HSI classification [40]. Unlike conventional band selection, spectral partitioning is a multiview spectral processing method that uses all spectral bands of the original HSI to generate its multiple views. Spectral partitions can be obtained by subsampling the original spectral bands, i.e., selecting an effective spectral subset to represent the original spectrum while retaining its main characteristics. In the field of HSI classification, there are many spectral partitioning methods, such as [40], [41], [42], and [43]. Liu et al. [42] first employ adaptive affinity propagation (AAP) [44] to automatically generate several clusters with correlated spectral bands and then perform a band reassignment to produce multiview band partitions of the original HSI. In [40], a spectrometer-driven (SD) spectral partitioning method is presented, which takes into account the physical properties of the spectrometer, while, in unmixing, there are also some attempts in this regard. In [43], unmixing is first attempted on a multiview dataset, and several abundance fusion methods are proposed. Recently, a spatial–spectral multiview data generation technique is presented in [45], and a multiview dictionary learning algorithm is proposed to tackle such data. Furthermore, Qi et al. [46] explore the application of multiview spectral information in sparse unmixing and propose a multiview collaborative sparse unmixing algorithm, which greatly improves the accuracy and robustness of HU.

Motivated by multiview unmixing, in this article, we explore and discuss the efficient application of multiview spectral and spatial information in AE-based unmixing models. Meanwhile, we propose a new multiview spatial–spectral two-stream network, called MSSS-Net, which collaboratively exploits the spatial and spectral information using cascaded RNNs in an end-to-end manner. The main contributions of this work can be summarized as follows.

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1) We explore the effective utilization of multiview spectral information, as well as neighborhood spatial information, in unmixing networks through spectral partitioning techniques. In the field of AE-based unmixing, we introduce the idea of multiview unmixing for the first time. We propose a novel multiview spatial–spectral two-stream deep unmixing network to learn deep patch-pixel features in HSI. It contains a spatial stream and a multiview spectral stream and shares a decoder, which is collaboratively trained in an end-to-end fashion.
2) We design a new cascaded bidirectional and unidirectional RNNs' encoder based on the long short-term memory (LSTM) to fully utilize the multiview spectral information of a single pixel and extract the deep spectral features of pixels. In the multiview spectral stream, the multiview spectral partitions of pixels are cyclically fed into a bidirectional RNN. The multi-input–multi-output characteristics of RNN are used to acquire the final fusion abundances, so as to improve the accuracy of the final estimated abundances.
3) The proposed cascaded RNNs' encoder can be perfectly suitable for the use of spatial information, which only needs to adjust the input and output forms. The spatial stream network can fully utilize the spatial information between pixels and their neighbors and extract deep spatial patch features. It loops the pixels in a patch into a bidirectional RNN and adopts the multi-input–single-output characteristics of RNN to obtain the abundance of the center pixel. High-quality endmembers are finally acquired through the reconstruction process of the AE network.
4) Our proposed MSSS-Net can efficiently analyze mixed spectra and fully explore bidirectional spectral correlations. In the field of unmixing, it is an effective alternative to CNNs and provides a perspective for constructing spatial–spectral unmixing networks using only RNNs.

The rest of this article is organized as follows. Section II introduces the background of the AE-based unmixing framework. In Section III, we detail the proposed MSSS-Net and its corresponding two-branch networks. Section IV discusses several experimental results and detailed comparisons with state-of-the-art networks. Finally, this article is concluded in Section V.

II. AE-Based Unmixing Framework

According to the LMM, an observed spectra matrix $Y \in R^{l\times n}$ with l-bands and n pixels can be expressed as

$$Y = \mathrm{A}\mathrm{X} + N \tag{1}$$

where $A \in R^{l\times p}$ represents an endmember spectra matrix with p endmembers, $X \in R^{p\times n}$ denotes the corresponding abundance coefficients, and $N \in R^{l\times n}$ is a noise matrix. Meanwhile, the abundance vector x_j for the jth pixel is required to satisfy the following two physical constraints, namely, ANC and ASC:

$$\begin{cases} x_j \geq 0 \\ \sum_{i=1}^{p} x_{ij} = 1. \end{cases} \tag{2}$$

The AE-based unmixing model simultaneously computes the endmembers A and abundances X in (1) via an unsupervised fashion. A conventional AE is generally designed as a three-layer structure and composed of an encoder and a decoder. The encoder encodes the spectra of mixed pixels $\{y_i\}_{i=1}^{n} \in R^l$ into a low-dimensional hidden representation h_i, which can be expressed as

$$h_i = f(y_i) = f\left(M^T y_i + b\right) \tag{3}$$

where f indicates an activation function and M represents the trainable parameters of encoder with b being its bias. In unmixing, the output of encoder h_i is generally regarded as the computed abundances, i.e., $\widehat{x}_i \leftarrow h_i$.

On the other hand, the decoder uses a hidden layer to reconstruct the spectra of mixed pixels from the code h_i and is expressed as

$$\hat{y}_i = g(h_i) = W^T h_i \tag{4}$$

where $\widehat{y}_i$ is the reconstructed spectrum of a mixed pixel and W represents the weight matrix of the decoder or the connection between hidden and output layers. For unmixing, the weight of decoder W is familiarly considered as the computed endmembers, i.e., $\hat{A} \leftarrow W$. From (3) and (4), we can observe that the encoder and decoder models naturally match with the unmixing task.

Given the AE-based unmixing architecture, the reconstruction error in the training process is mainly implemented by minimizing the mean square error (mse)-based loss function, which is formed as

$$L_{\mathrm{AE}} = \frac{1}{n}\sum_{i=1}^{n} \|\widehat{y}_i - y_i\|^2. \tag{5}$$

It is worth noting that, in the AE-based unmixing framework, the bias b is generally not used because it tends to be large negative values [26].

III. Multiview Spatial–Spectral Two-Stream Network

In order to fully utilize the multiview spectral information and spatial information in HSI, we propose a novel multiview spatial–spectral two-stream unmixing network. Based on the finding in [39] that spatial information is beneficial for endmember extraction, while spectral information is more effective for abundance estimation, we adopt a spatial–spectral two-stream network framework. It consists of a patch spatial information-oriented spatial RNN and a multiview spectral information-oriented multiview spectral RNN. The two-stream network estimates endmembers and abundances collaboratively through a weight-sharing strategy. Fig. 1 presents the proposed MSSS-Net framework.

Meanwhile, to fully explore the specific spatial–spectral properties of mixed pixels, we design a cascaded RNNs model using LSTMs [47] to capture discriminative depth patch-pixel features. It mainly contains two RNN structures. The first structure is a bidirectional RNN that captures complementary information from different spectral bands by learning bidirectional spectral correlations. The second structure is a

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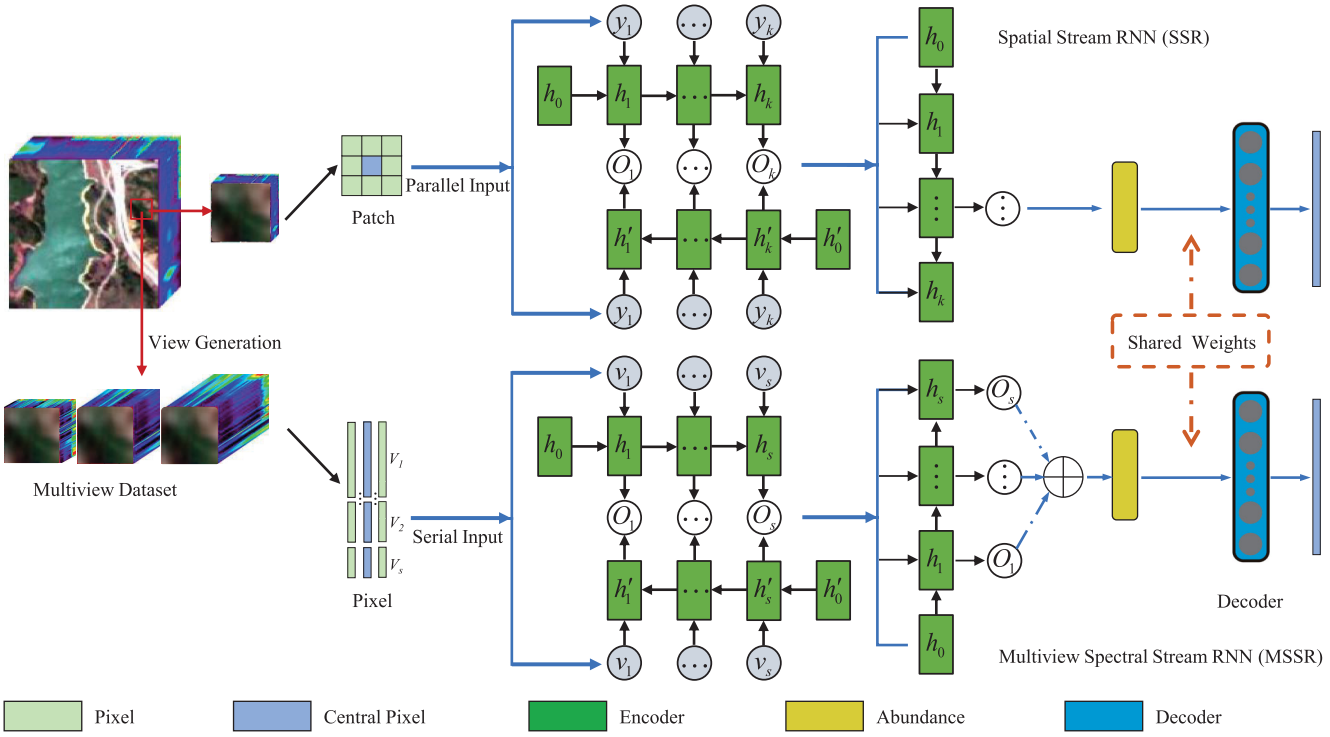


Fig. 1. Network architecture of the proposed MSSS-Net.

unidirectional RNN that learns from fused features to acquire a more accurate representation of abundance. The two-stream unmixing network uses the same cascaded RNNs' encoder with different input data and input–output forms to exploit spatial and multiview spectral information, respectively.

In the spatial stream RNN, referred to as SSR, we propose a patch-based recurrent perception method for spatial information. In traditional unmixing approaches, the region-based processing methods have attained good results [45], [48], [49], among which the sliding window-based methods are a typical representative [18], [46], [50]. Such methods can extract good endmembers since it supports the idea of endmember bundles [30], [51]. Thus, we integrate the popular patch-based method into the RNN encoder to explore spatial information in sequences of mixed spectra. It adopts a parallel bidirectional multi-input–single-output way to obtain the representation of center pixels. Finally, endmembers are obtained by reconstructing the center pixels with a linear decoder.

In the multiview spectral stream RNN, MSSR for short, we propose a spectral complementary recurrent fusion method for multiview spectral information. Among deep learning-based unmixing approaches, we observe that vector-based 1-D CNNs are widely adopted to process spectral information [52], [53], [54]. Using the convolution operation to process the whole spectral bands of a single pixel does generate decent abundances [53]. However, it may lead to information loss due to the intrinsic sequence-based data structure of mixed pixels [55], [56]. In addition, the spectral bands of mixed pixels have redundant information, which increases the difficulties of capturing complementary features [57], [58]. Based on our previous work [46], we introduce multiview spectral information and integrate it into the RNN encoder to fully exploit the potential of narrow spectral bands in HSI. It adopts a serial bidirectional multi-input–multi-output way to acquire multiview abundance coefficients and obtains more accurate abundances through a fusion RNN.

Note that both SSR and MSSR are AE-based unmixing networks employing cascaded RNNs, which exploits the spatial and spectral information of HSI. They can be used as independent unmixing networks that estimate endmembers and abundances simultaneously. Based on our previous findings [39], we adopt a weight-sharing collaborative strategy to take full advantage of the respective strengths of SSR and MSSR. In addition to improving the accuracy of endmember and abundance estimation, this strategy can raise the convergence speed of the proposed network.

In the following, we will describe in detail the SSR and MSSR, and the multiview data construction method.

A. Spatial Stream RNN

Patch- or region-based methods are widely used to deal with spatial information in HSI. It is based on the assumption that the spectral features of adjacent pixels are highly correlated and share a set of endmembers, and therefore, the distribution of the same endmember is continuous and smooth [18], [23]. Huang et al. [50] introduce spatial information into the AE-based unmixing model through a patch-based strategy. However, like all traditional methods, the spatial weights are still fixed based on a certain spatial model. We, therefore, propose a cascaded RNNs' encoder to further exploit the potential of spatial information in a patch. Differently, in the

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SSR network, the spatial information among pixels in a patch is recurrently perceptional through a bidirectional RNN and a unidirectional RNN.

For any patch P, which contains k mixed pixels, we denote its center pixel as y_c. In the SSR network, we feed all mixed pixels in a patch into the bidirectional RNN implemented by LSTMs in a parallel way, as shown in the top half of Fig. 1. Such two layers of LSTMs model the forward and backward spatial correlations of pixels within the provided patch. We adopt a many-to-many manner to generate the hidden representations h_t and h'_t of each pixel. These two feature representations are further concatenated to an activation layer as the output O_t of bidirectional LSTMs on pixel y_t, which can be expressed as

$$O_t = \varphi\left(h_t \oplus h'_t\right), \quad t = 1,\ 2, \ldots, k \tag{6}$$

where $\varphi(\cdot)$ is the activation function, of which we adopt the SELU function in this article. Here, $\oplus$ denotes the matrix splicing, and the activations h_t and h'_t of the forward and reverse LSTM units can be calculated by

$$h_t = o_t \tanh(c_t) \tag{7}$$

$$h'_t = o'_t \tanh\left(c'_t\right) \tag{8}$$

where $\tanh(\cdot)$ is the hyperbolic tangent function of the system state. The memory cell c_t on pixel y_t and three "gates" of the LSTM unit including an output gate o_t, an input gate i_t, and a forget gate f_t are given by

$$o_t = \sigma\left(W_{\mathrm{ho}} h_{t-1} + W_{\mathrm{yo}} y_t + b_o\right) \tag{9}$$

$$c_t = c_{t-1} \odot f_t + \widetilde{c}_t \odot i_t \tag{10}$$

$$\widetilde{c}_t = \tanh\left(W_{\mathrm{hs}} h_{t-1} + W_{\mathrm{ys}} y_t + b_s\right) \tag{11}$$

$$i_t = \sigma\left(W_{\mathrm{hi}} h_{t-1} + W_{\mathrm{yi}} y_t + b_i\right) \tag{12}$$

$$f_t = \sigma\left(W_{\mathrm{hf}} h_{t-1} + W_{\mathrm{yf}} y_t + b_f\right) \tag{13}$$

where $\sigma(\cdot)$ is the sigmoid function, $\widetilde{c}_t$ is the updated component of c_t, $\odot$ represents an elementwise product, W terms with different subscripts denote weight matrices with distinct data-flow paths, and b are bias coefficients. Likewise, the computation of h'_t is the same.

Then, O_t is fed into the subsequent unidirectional LSTM unit. After performing further recurrent perception on the concatenated feature representations, we obtain the low-dimensional hidden representation h''_c of the center pixel in a many-to-one fashion. Meanwhile, the output O'_c of unidirectional LSTM on the center pixel is obtained through an activation layer. This transformation can be described as

$$O'_c = \varphi\left(h''_c\right) \tag{14}$$

$$h''_c = o''_c \tanh\left(c''_c\right) \tag{15}$$

where h''_c can be solved using (9)–(13) in the same way as h_t. Note that the activation function $\varphi(\cdot)$ here is replaced by the RELU function.

After encoding the pixels in an input patch through the cascaded RNNs, according to (3), O'_c is the estimated abundance of the center pixel y_c. We employ the l_1 norm to impose the ASC constraint on the estimated abundance to obtain a better reconstruction result. The final abundance of center pixel y_c in a patch is denoted as

$$x_c = \frac{O'_c}{\|O'_c\|_1 + \varepsilon} \tag{16}$$

where ε is a small constant.

Finally, y_c is reconstructed by a linear decoder, which is expressed as

$$\hat{y}_c = W_d x_c \tag{17}$$

where the weights W_d of the decoder are the finally estimated endmembers.

The previous experience [29], [34], [39] confirms the excellent properties of spectral angle distance (SAD), so we employ SAD as the underlying loss in this article. Meanwhile, considering the sparsity of abundance fractions, we apply the $l_{1/2}$ norm [12] to the abundance vector for a better reconstruction. Therefore, the loss function of the SSR network is written as

$$L_{\mathrm{SSR}} = \frac{1}{n} \sum_{c}^{n} \left(\mathrm{SAD}(y_c, \hat{y}_c) + \lambda \|x_c\|_{\frac{1}{2}}\right). \tag{18}$$

TABLE I

STRUCTURE OF THE PROPOSED CASCADED RNNS MODULE

Settings	Bidirectional LSTM	LSTM for SSR	LSTM for MSSR
Dropout	√		
Batch Normalization	√	√	√
Activation Function	SELU	RELU	Sigmoid

B. Multiview Data Construction

The multiview dataset is generated from the original HSI by spectral partitioning. As we introduced in [46], we mainly consider two typical partitioning techniques, i.e., AAP [42] and SD [40], which are based on statistical and physical principles, respectively, and can be easily applied to different HSI types.

AAP [44] is an efficient clustering algorithm that can cluster spectral bands, while the number of samples is unknown. Based on [42], after performing AAP partitioning on an HSI, r-band clusters are obtained, which includes a set of highly correlated spectral bands. Considering the specifications of the imaging spectrometer itself, Liu et al. [40] propose an SD-based band partitioning method. Different components of the spectrometer correspond to a set of highly correlated spectral bands, which can also be regarded as the basis for spectral partitioning. Let q represent the number of components in a spectrometer, and each component corresponds to a spectral group containing known spectral wavelengths.

After executing the AAP or SD strategy, r- or q-band clusters (spectral groups) are acquired. We then employ the uniform band sampling introduced in [40] to reassign similar spectral bands into s partitions. It is worth pointing out that each spectral partition embodies bands from all band clusters (spectral groups) with a low correlation, which is determined by s. Therefore, each view subset of the multiview dataset generated by spectral partitioning maintains the basic shape of

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the original HSI. In other words, the spectral signatures in each view subset preserve the same spectral characteristics as the original HSI. This sampling strategy reduces the dimensionality and redundancy of the original data and guarantees that the same endmembers are included in different view subsets. Finally, s views of the raw hyperspectral data $\{Y^{\upsilon}\}_{\upsilon=1}^{s}$ can be obtained after spectral partitioning. Meanwhile, $Y^{\upsilon} \in R^{L^{\upsilon}\times n}$ represents the υth view of the raw data, which has L^{υ}-bands and n pixels. From another perspective, different view subsets possess the same raw mixed pixels but with distinct spectral bands.

C. Multiview Spectral Stream RNN

Unlike existing unmixing algorithms, which process the entire spectrum of the mixed pixel at once, we introduce the multiview dataset and incorporate it into the proposed cascaded RNNs' encoder. The bidirectional LSTMs in the cascaded RNNs' encoder can recurrently perceive spectral sequences and capture spectral complementary information, while the unidirectional LSTM can cyclically fuse abundance features to improve the estimated abundances.

Different from the patch-based parallel input manner of SSR, MSSR performs pixel-based multiview unmixing, that is, the input of MSSR is the spectral partitions of mixed pixels, as shown in the bottom half of Fig. 1. After constructing the multiview dataset, we can obtain s views data subsets $\{y_c^{\upsilon}\}_{\upsilon=1}^{s}$ for any pixel y_c. The s spectral partitions of y_c are fed into the bidirectional LSTMs to bidirectionally model the spectral complementarity information and spectral correlations between bands. We use a many-to-many manner to obtain the hidden representations h_c^{υ} and $h_c^{\upsilon'}$ for each spectral partition and generate the output O_c^{υ} of bidirectional RNNs over view υ through an activation layer. This process can be expressed as

$$O_c^{\upsilon} = \varphi\left(h_c^{\upsilon} \oplus h_c^{\upsilon'}\right), \quad \upsilon = 1,\ 2, \ldots, s \tag{19}$$

where $\varphi(\cdot)$ is the SELU function. Similarly, h_c^{υ} and $h_c^{\upsilon'}$ of the forward and reverse LSTM units can be computed by

$$h_c^{\upsilon} = o_c^{\upsilon} \tanh\left(c_c^{\upsilon}\right) \tag{20}$$

$$h_c^{\upsilon'} = o_c^{\upsilon'} \tanh\left(c_c^{\upsilon'}\right) \tag{21}$$

respectively, which can be solved in the same way as h_t.

In the subsequent unidirectional LSTM unit, we still adopt a many-to-many fashion to generate the low-dimensional hidden representation $h_c^{\upsilon''}$ of each view. Meanwhile, the output $O_c^{\upsilon'}$ is acquired through an activation layer as

$$O_c^{\upsilon'} = \varphi\left(h_c^{\upsilon''}\right) \tag{22}$$

$$h_c^{\upsilon''} = o_c^{\upsilon''} \tanh\left(c_c^{\upsilon''}\right). \tag{23}$$

It is worth noting that the activation function $\varphi(\cdot)$ here is the sigmoid function.

As in the SSR network, $O_c^{\upsilon'}$ is the estimated abundance of each view, and the final abundance x_c^{υ} can be obtained after applying the ASC constraint, which is given by

$$x_c^{\upsilon} = \frac{O_c^{\upsilon'}}{\|O_c^{\upsilon'}\|_1 + \varepsilon}. \tag{24}$$

The abundance x_c^{υ} is obtained by the recurrent perception of the concatenated feature representations under each view through the unidirectional LSTM unit. To generate the fused abundance, we take the average abundance under s views as the final abundance of pixel y_c, which is denoted as

$$x_c = \frac{1}{s}\sum_{\upsilon=1}^{s} x_c^{\upsilon}. \tag{25}$$

The decoder is shared with the SSR network, and the pixel y_c is reconstructed similarly as

$$\hat{y}_c = W_d x_c. \tag{26}$$

The loss function of the MSSR network also contains a reconstruction term and a sparse term as

$$L_{\mathrm{MSSR}} = \frac{1}{n}\sum_{c}^{n}\left(\mathrm{SAD}(y_c, \hat{y}_c) + \lambda\frac{1}{s}\sum_{\upsilon}^{s}\|x_c^{\upsilon}\|_{\frac{1}{2}}\right). \tag{27}$$

The network structure of the cascaded RNNs' encoder module is summarized in Table I. The proposed MSSS-Net adopts a two-stream network structure, where the decoders share a set of weights. This collaborative strategy can further explore the potential of the two branch networks. The endmembers estimated by the SSR network will have a positive effect on the MSSR network and drive it to a more interpretable abundance result. The overall loss function can be written as

$$L = \mu L_{\mathrm{SSR}} + (1-\mu)L_{\mathrm{MSSR}}. \tag{28}$$

IV. Experiments

In this section, we evaluate the performance of MSSS-Net on real and synthetic datasets. We consider three traditional unmixing methods: VCA [6], $L_{1/2}$-NMF [12], and Dgs-NMF [14], where VCA is a well-known baseline algorithm, $L_{1/2}$-NMF adopts the abundance sparsity, and Dgs-NMF considers spatial–spectral information. We also compare six state-of-the-art AE-based unmixing networks: EndNet [29], SNMF-Net [33], and TANet [30] only utilize spectral information of HSI, while CNNAEU [34], MiSiC-Net [38], and SSAE [50] are spatial–spectral-based unmixing networks.

A. Performance Metrics and Datasets

The SAD and root mse (RMSE) are used to evaluate the unmixing performances, which are defined as

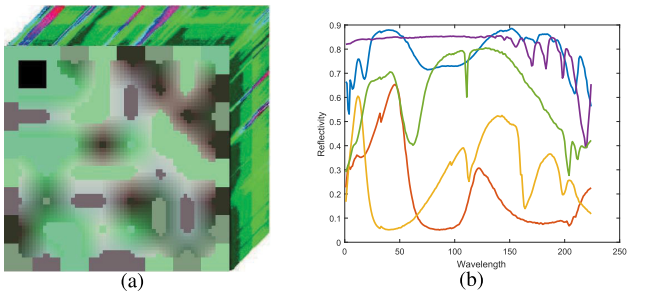
$$\mathrm{SAD}\left(a_c, \widehat{a}_c\right) = \arccos\left(\frac{a_c \cdot \widehat{a}_c}{\|a_c\|_2\|\widehat{a}_c\|_2}\right) \tag{29}$$

$$\mathrm{RMSE}\left(x_c, \widehat{x}_c\right) = \sqrt{\frac{1}{n}\sum_{c=1}^{n}\|x_c - \widehat{x}_c\|_2^2}. \tag{30}$$

It is worth noting that the smaller values of both metrics indicate better performances.

We use a serious of synthetic datasets and three real datasets to evaluate different unmixing methods, which are detailed as follows. The ground truths (GTs) of endmembers and abundances for the real datasets are determined via the methods presented in [11], [12], [14], and [59].

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(a) (b)

Fig. 2. (a) Synthetic dataset. (b) Endmembers in the dataset selected from the USGS spectral library.



(a) (b) (c)

Fig. 3. Cube images of three real datasets used in the experiments. (a) Urban. (b) Jasper Ridge. (c) Cuprite.

1) Synthetic Dataset: The synthetic dataset is provided by Tang et al. [17] and contains 64 × 64 pixels. It holds a certain spatial structure and exhibits spatial homogeneity, as shown in Fig. 2. The five endmembers in this dataset are chosen from the U.S. Geological Survey (USGS) spectral library [60], and the abundances meet the ANC and ASC. Meanwhile, the dataset is corrupted by independent identically distributed Gaussian noise with different levels of signal-to-noise ratio (SNR).

2) Real Dataset 1 (Urban): The Urban dataset is collected by the HYDICE sensor and includes 307 × 307 pixels. After removing low-quality channels, the remaining 162 bands are used for unmixing. Fig. 3(a) shows a cube image of the Urban dataset, which consists of four constituent materials: Asphalt, Grass, Tree, and Roof.

3) Real Dataset 2 (Jasper Ridge): The Jasper Ridge dataset is obtained with the AVIRIS sensor and contains 100×100 pixels. Similarly, after removing low-quality bands, 198 channels are used for unmixing. Fig. 3(b) shows a cube image of the Jasper Ridge dataset, which has four main endmembers: Tree, Water, Soil, and Road.

4) Real Dataset 3 (Cuprite): The Cuprite dataset is captured with the AVIRIS sensor over Cuprite, NV, USA, and includes 250 × 190 pixels. After removing the noisy and water absorption channels, the remaining 188 bands are used for unmixing. Fig. 3(c) presents a cube image of the dataset, which consists of 12 unique mineral endmembers: Alunite, Andradite, Buddingtonite, Dumortierite, Kaolinite1, Kaolinite2, Muscovite, Montmorillonite, Nontronite, Pyrope, Sphene, and Chalcedony.

B. Experiments on MSSS-Net

1) Parameters and Complexity Analysis: The MSSS-Net is implemented under the environment of Python 3.7 and PyTorch 1.0 and trained on CPUs. The Adam optimizer is adopted during training. The learning rate is set to $1e-4$ for the cascaded RNNs' encoder, and the number of epochs is set to 100. The tradeoff parameter λ is empirically set to $5e-5$. Considering that SSR and MSSR have the same contribution to the whole network, μ is set to 0.5. The endmember weights of these networks are (partial) initialized from the endmembers extracted by the VCA algorithm.

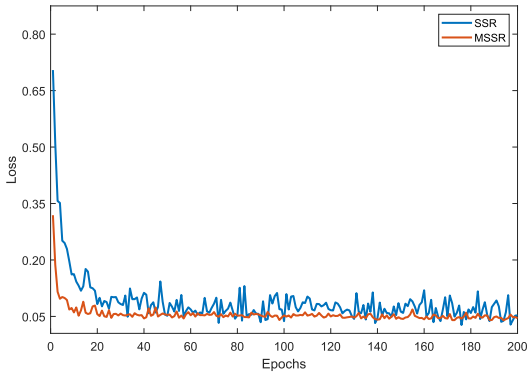


Fig. 4. Loss as a function of epochs for the SSR and MSSR in the proposed MSSS-Net.

TABLE II
COMPLEXITY OF THE AE-BASED UNMIXING NETWORKS

Networks	EndNet	SNMF-Net	TANet	CNNAEU	MiSiCNet	SSAE	MSSS-Net
#Params	1.0k-1.5k	13k	15k	1.0k-1.5k	1.5M	20k	13k

In addition, the acquired loss functions for SSR and MSSR on the synthetic dataset with 30-dB SNR are given in Fig. 4. The number of iterations is set to 200. We can observe that the convergence curves of SSR and MSSR are gratifying. Both SSR and MSSR networks can converge in a relatively small number of iterations. This benefits from the adopted weight-sharing collaborative strategy. It enables the SSR and MSSR networks to correct and promote each other and boosts the convergence speed of the whole network.

Finally, the computational complexity is also a significant parameter for the AE network-based unmixing algorithms, which affects the running time and efficiency of the unmixing methods. In Table II, we calculate and list the number of parameters of the seven unmixing algorithms based on the AE network. It can be seen that the number of parameters of all unmixing networks is not large since the present unmixing network architectures are comparatively simple. Meanwhile, our proposed MSSS-Net has fewer parameters than TANet, MiSiCNet, and SSAE. This is because MSSS-Net is a spatial–spectral unmixing network using only RNNs.

2) Spatial Information Analysis: In the SSR network, the cascaded RNNs' encoder performs recurrent perception on mixed pixels in a given patch to fully exploit the spatial information contained in the patch. Accordingly, the patch size becomes a key parameter. Fig. 5 shows the variation of the average SAD values of extracted endmembers with patch size on different datasets, where each experimental result is performed 20 times, and the average value is reported. The patch size of 1 × 1 corresponds to a single mixed pixel, i.e., the case where spatial information is not considered.

From Fig. 5, we can see that, on all datasets, as the sliding window becomes larger, the endmember extraction performance is improved first but then decreased. The best unmixing

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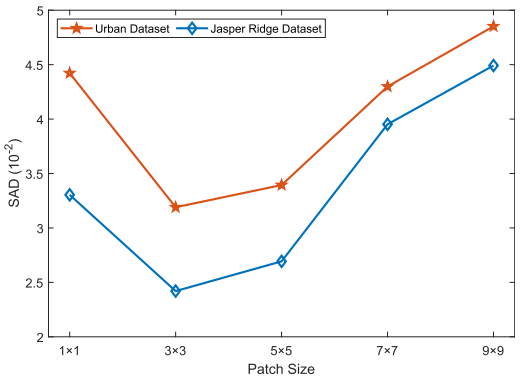


Fig. 5. Performance of the proposed MSSS-Net with respect to different patch sizes.

results are concentrated around patch sizes of 3×3 and 5×5. The original intention of our method is to estimate the abundance of a center pixel by perceiving the center pixel and its neighbors. However, as the patch size becomes larger, the differences between the neighboring pixels and the center pixel become larger, and the actual abundance differences are also large. If such abundances with large differences are cyclically sensed with the abundance of the central pixel, it will lead to a decrease in the accuracy of abundance estimation and ultimately affect the extracted endmembers.

Meanwhile, it can be found from the curves that the optimal value of patch sizes is 3×3, which is consistent with the results in [35], [50], and [51]. The difference is that we fully mine the spatial information in a given patch through a cascade RNNs' encoder. It captures the spatial correlations between mixed pixels in a loop-aware manner to extract better endmembers. Furthermore, we can also see that different patch sizes have different effects on different datasets. On the Urban dataset, a 7×7 window can still achieve an improvement compared to the case where no spatial information is used, while, on the Jasper Ridge dataset, a 7×7 window is not a good choice. It is related to the spatial homogeneity of the dataset itself. The Urban dataset is larger and has a higher spatial resolution, which makes it richer in spatial information available.

3) Multiview Spectral Strategy Analysis: In the MSSR network, we introduce multiview spectral information through spectral partitioning techniques. The cascaded RNNs' encoder recurrently fuses spectra from different views of a given pixel to fully exploit the spectral information contained in the pixel. How to choose the spectral partitioning strategy and the optimal number of views is the key. Fig. 6 presents the mean RMSE values of estimated abundances under different partitioning strategies as a function of the number of views, i.e., s, from 1 to 9. The number of views $s = 1$ corresponds to the single view case, where the multiview spectral information is not considered.

From Fig. 6(a) and (b), we can see that the results under the two spectral partitioning strategies show the same trend. On all datasets, the abundance estimation effect increases correspondingly as the number of views increases and then begins to decrease. Meanwhile, the best unmixing results are around $s = 2$ and $s = 3$. The spectral partitioning technique can reduce the spectral similarity between bands and increase the diversity of partitions. However, as the number of views (partitions) increases, the spectral bands contained in each view decrease, which affects the capture and learning of spectral information by the network. Therefore, there is an equilibrium value for the number of views, which is around $s = 2$ or $s = 3$. It is worth pointing out that, with the development of remote sensing and imaging spectrometer technology, the spectral resolution of acquired images becomes finer, and the number of bands is increasing, such as in ultraspectral images. Under this trend, it is foreseeable that multiview spectral unmixing will exhibit greater potential.

On the other hand, it can be found from the histograms that the SD spectral partitioning strategy is slightly better than AAP in most cases, thanks to its physical model-based grouping strategy. Furthermore, we can also find that different numbers of views perform differently on different datasets. On the Urban dataset, when $s = 4$, the unmixing effect is not as good as that of the single view case, where multiview spectral information is not used, while, on the Jasper Ridge dataset, an improvement can still be obtained under $s = 4$. It is still related to the characteristics of the dataset itself. The Jasper Ridge dataset contains more spectral bands, which can provide more bands for the data subset of each view under the same number of views. Based on the above experimental results, the experiments in this article adopt the SD partitioning strategy and set $s = 2$.

C. Comparison of MSSS-Net Against Other Methods

1) Experiments on Synthetic Dataset: We investigate the robustness of the proposed MSSS-Net on the synthetic dataset under different noise levels, which are set as 20, 30, 40, 50, and 60 dB. The endmember and abundance estimation results under different SNRs are shown in Fig. 7(a) and (b), including the results of all the alternative unmixing methods. Due to the poor unmixing performance of VCA and $L_{1/2}$-NMF, we do not include their unmixing results in this experiment.

In general, the mean SAD and RMSE values obtained by all the unmixing methods show a decreasing trend as the SNR increases. From Fig. 7, we can see that most AE-based unmixing networks achieve better results than traditional unmixing methods, except for CNNAEU. CNNAEU provides inferior SAD and RMSE values, which may be related to the size of the synthetic dataset. Moreover, it can be seen that MiSiCNet, SSAE, and MSSS-Net based on spatial–spectral information outperform spectral-based EndNet, SNMF-Net, and TANet, especially in low-quality HSIs. Among them, the proposed MSSS-Net provides more desirable unmixing results, which have comparable advantages under all SNR values. This is mainly due to its more efficient feature representation capability based on the RNN encoder. It also attributes to the proposed two-stream collaborative network architecture, which simultaneously possesses patch-based spatial recurrent perception and multiview spectral-based recurrent fusion capabilities.

2) Experiments on Urban Dataset: Table III presents the endmember results extracted by different unmixing methods

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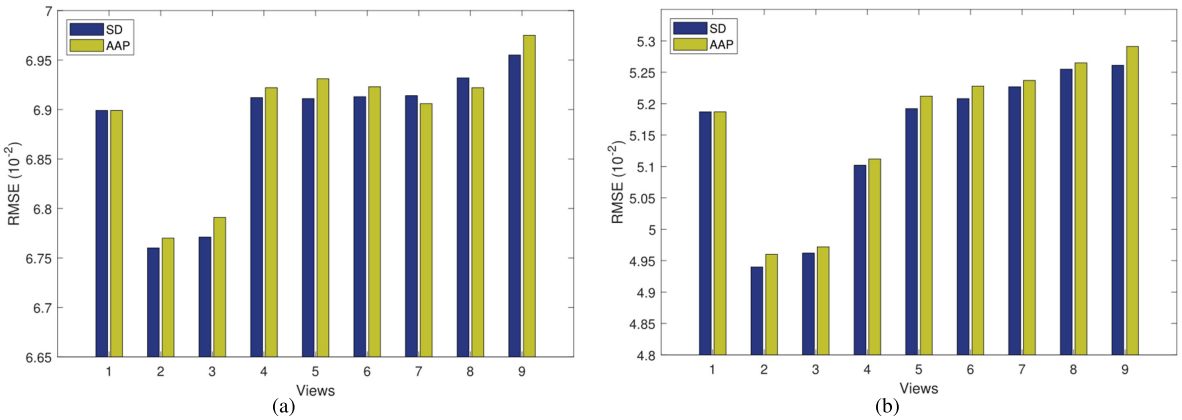


Fig. 6. Experimental results of different spectral partitioning strategies. (a) Results on the Urban dataset. (b) Results on the Jasper Ridge dataset.

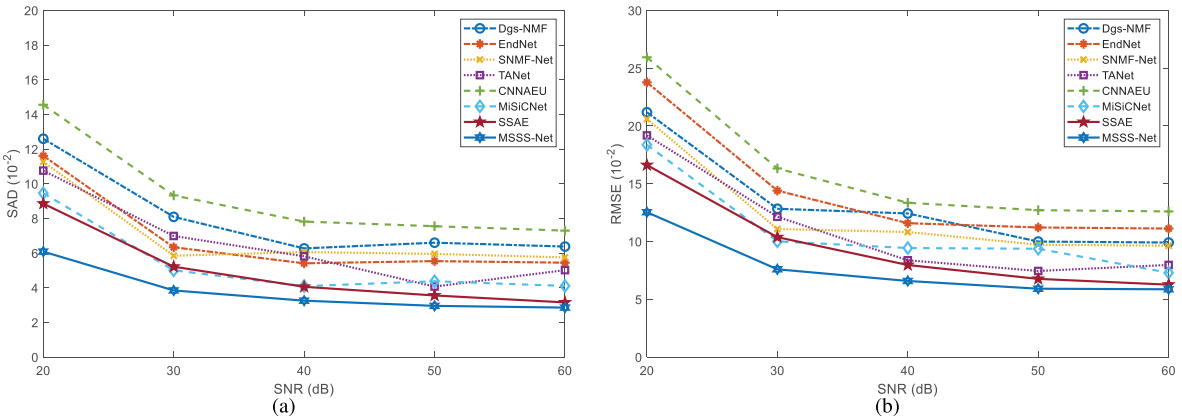


Fig. 7. Experimental results of different unmixing methods with respect to different noise levels. (a) Mean SAD values. (b) Mean RMSE values.

TABLE III

ENDMEMBER EXTRACTION RESULTS ($\times 10^{-2}$) BY DIFFERENT UNMIXING METHODS ON THE URBAN DATASET. THE BEST RESULTS ARE IN BOLD

Endmember	VCA	$L_{1/2}$-NMF	Dgs-NMF	EndNet	SNMF-Net	TANet	CNNAEU	MiSiCNet	SSAE	MSSS-Net
Asphalt	18.85 ± 7.2	6.06 ± 0.2	5.86 ± 0.1	5.86 ± 0.2	6.21 ± 0.5	3.85 ± 0.3	5.75 ± 0.5	**3.84 ± 0.6**	6.19 ± 0.5	4.86 ± 0.2
Grass	33.48 ± 13.0	20.05 ± 0.4	13.69 ± 0.4	5.36 ± 0.1	5.20 ± 0.8	3.66 ± 0.3	3.66 ± 0.4	7.51 ± 0.8	3.43 ± 0.3	**3.13 ± 0.1**
Tree	18.35 ± 13.0	3.71 ± 0.0	4.12 ± 0.0	4.68 ± 0.2	4.37 ± 0.8	7.49 ± 0.4	3.21 ± 0.3	3.69 ± 0.4	**2.95 ± 0.2**	3.07 ± 0.2
Roof	71.42 ± 0.2	14.22 ± 0.2	10.54 ± 0.3	3.89 ± 0.7	4.61 ± 0.5	2.50 ± 0.3	3.32 ± 0.6	1.75 ± 0.2	2.97 ± 0.3	**1.69 ± 0.1**
Average	35.52 ± 4.1	11.01 ± 0.2	8.85 ± 0.2	4.95 ± 0.3	5.10 ± 0.5	4.38 ± 0.3	3.99 ± 0.3	4.20 ± 0.4	3.89 ± 0.1	**3.19 ± 0.1**

on the Urban dataset, where each experimental result is performed 20 times to obtain the mean and standard deviation of SAD. We can see that the error of endmembers extracted by the geometry-based VCA is large. The NMF-based methods have achieved a good improvement compared with the VCA, and Dgs-NMF considering spatial information outperforms $L_{1/2}$-NMF that only uses spectral information, which shows the importance of spatial information. While SNMF-Net, which combines model-based and learning-based methods, achieves the best endmembers among NMF-based approaches. The remaining six AE-based unmixing networks have all achieved ideal results, which are 45%–65% higher than traditional methods, demonstrating the superiority of the AE-based structure. Specifically, TANet estimates better endmembers than EndNet, thanks to its additional endmember guidance network. The four spatial–spectral-based unmixing networks, i.e., CNNAEU, MiSiCNet, SSAE, and MSSS-Net, naturally achieve better endmember estimation than the spectral-based unmixing networks (TANet and EndNet). Among them, MSSS-Net can extract the best endmembers, which shows the advantage of RNN compared to CNN in unmixing. Meanwhile, compared with SSAE, MSSS-Net can make better use of spatial information through recurrent perception to obtain better unmixing results.

Table IV shows the abundance estimation results for the compared unmixing methods. The abundances obtained by VCA are also rough due to inaccurate endmembers. The abundances estimated by $L_{1/2}$-NMF and Dgs-NMF considering sparse and spatial information are greatly improved. Similarly, the seven unmixing networks based on AE all achieve decent abundances. The difference is that the abundances estimated by EndNet and CNNAEU do not compare favorably with

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TABLE IV

ABUNDANCE ESTIMATION RESULTS ($\times 10^{-2}$) BY DIFFERENT UNMIXING METHODS ON THE URBAN DATASET. THE BEST RESULTS ARE IN BOLD

Abundance	VCA	$L_{1/2}$-NMF	Dgs-NMF	EndNet	SNMF-Net	TANet	CNNAEU	MiSiCNet	SSAE	MSSS-Net
Asphalt	33.05 ± 19.7	14.77 ± 0.1	13.18 ± 0.1	10.63 ± 0.1	9.36 ± 0.4	**7.05 ± 0.3**	12.49 ± 0.3	9.71 ± 0.4	9.98 ± 0.4	8.07 ± 0.3
Grass	46.56 ± 7.9	16.16 ± 0.2	12.95 ± 0.0	13.86 ± 0.3	11.15 ± 0.5	12.07 ± 0.4	12.57 ± 0.3	9.94 ± 0.3	9.75 ± 0.2	**8.96 ± 0.2**
Tree	31.32 ± 20.2	12.65 ± 0.2	9.57 ± 0.1	9.17 ± 0.3	8.76 ± 0.3	6.34 ± 0.4	8.55 ± 0.2	6.92 ± 0.2	5.68 ± 0.2	**5.00 ± 0.1**
Roof	17.25 ± 3.2	6.90 ± 0.1	6.27 ± 0.0	6.40 ± 0.2	10.44 ± 0.2	9.78 ± 0.3	8.54 ± 0.2	6.42 ± 0.2	5.65 ± 0.2	**5.01 ± 0.2**
Average	32.04 ± 2.4	12.62 ± 0.1	10.49 ± 0.1	10.02 ± 0.2	9.92 ± 0.4	8.81 ± 0.3	10.54 ± 0.1	8.25 ± 0.2	7.77 ± 0.1	**6.76 ± 0.1**

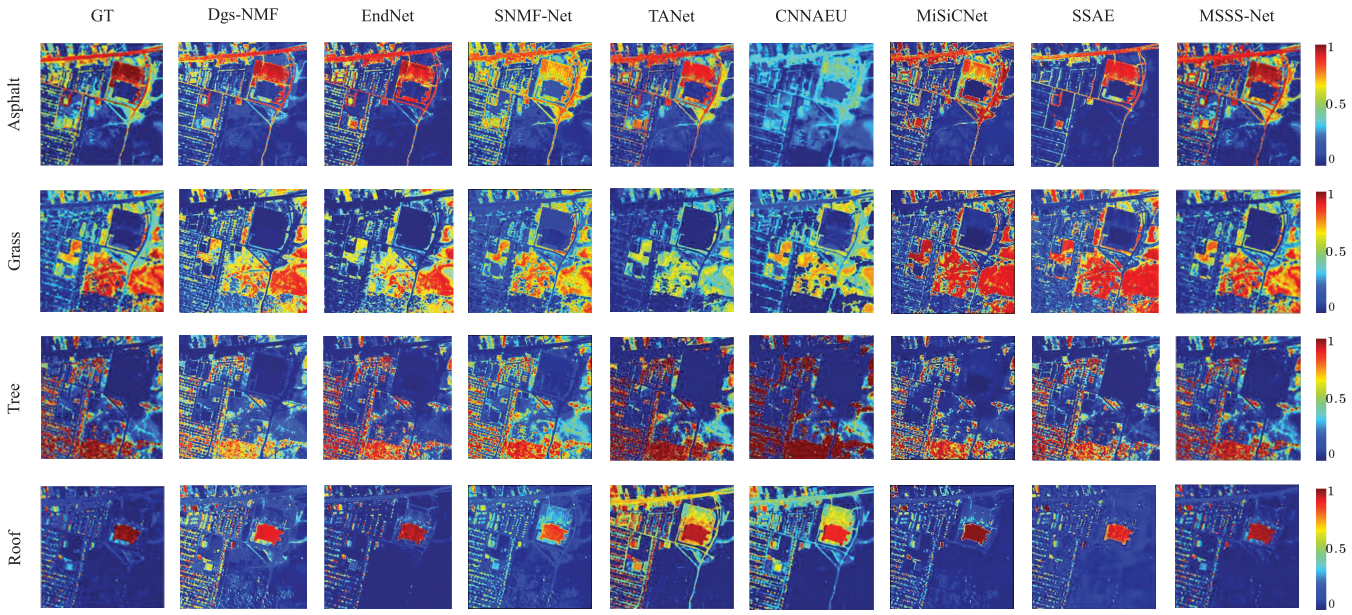


Fig. 8. Abundance maps estimated by different unmixing methods on the Urban dataset and the corresponding GTs.

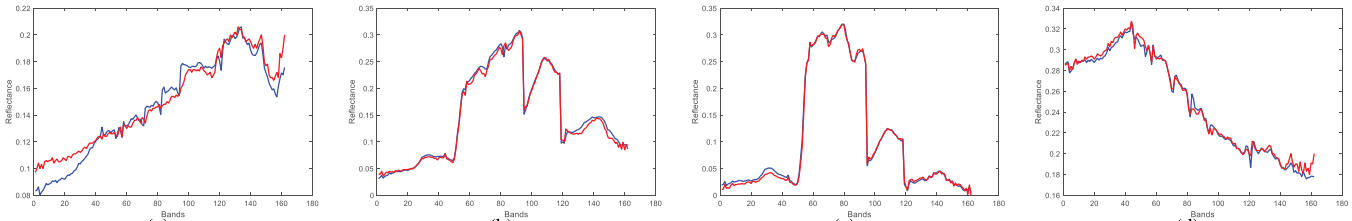


Fig. 9. Endmembers extracted by MSSS-Net (blue curves) on the Urban dataset and the corresponding GTs (red curves). (a) Endmember Asphalt. (b) Endmember Grass. (c) Endmember Tree. (d) Endmember Roof.

traditional NMF-based methods. The TANet with the two-stream network structure offers a certain improvement on abundances and achieves the best estimation result on endmember Asphalt. The abundances of SSAE and MiSiC-Net using the convolutional AE network have been greatly improved, and they have more than 20% improvement compared to EndNet and CNNAEU. Ultimately, MSSS-Net achieves the best abundance results through the recurrent fusion strategy. This confirms the superiority of RNN in abundance estimation and the advantage of the two-stream network.

For illustrative purposes, Fig. 8 visualizes all the abundance maps of main comparison methods, which are qualitative representations of the abundances in Table IV. Corresponding to the abundance values in Table IV, the abundance maps of MiSiCNet, SSAE, and MSSS-Net are all close to the GT, while MSSS-Net shows more details. Meanwhile, Fig. 9 visualizes the endmember spectral curves extracted by MSSS-Net. It can be seen that the endmembers Grass, Tree, and Roof completely coincide with GTs. This fully demonstrates the superiority of cascaded RNN encoders.

3) Experiments on Jasper Ridge Dataset: Table V presents the endmember results estimated by different unmixing methods on the Jasper Ridge dataset. Similarly, $L_{1/2}$-NMF and Dgs-NMF considering sparse and spatial information perform better than the geometry-based VCA. SNMF-Net still has advantages over NMF-based methods. The AE-based unmixing networks achieve ideal estimation results, except for CNNAEU. The reason for the poor performance of CNNAEU on the Jasper Ridge dataset may be related to the size

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TABLE V

ENDMEMBER EXTRACTION RESULTS ($\times 10^{-2}$) BY DIFFERENT UNMIXING METHODS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE IN BOLD

Endmember	VCA	$L_{1/2}$-NMF	Dgs-NMF	EndNet	SNMF-Net	TANet	CNNAEU	MiSiCNet	SSAE	MSSS-Net
Tree	13.95 ± 3.1	15.10 ± 0.3	4.66 ± 0.2	4.58 ± 0.4	4.52 ± 1.2	2.85 ± 0.2	11.94 ± 2.1	3.45 ± 0.2	3.38 ± 0.1	**2.56 ± 0.1**
Water	29.09 ± 10.2	4.60 ± 0.0	4.60 ± 0.0	5.06 ± 0.9	5.30 ± 1.5	3.28 ± 0.3	6.92 ± 0.4	6.35 ± 0.3	4.96 ± 0.1	**3.06 ± 0.1**
Soil	15.22 ± 2.3	6.16 ± 0.5	5.66 ± 0.2	5.17 ± 0.3	4.26 ± 1.3	8.50 ± 0.4	10.15 ± 0.9	3.31 ± 0.1	3.28 ± 0.3	**1.80 ± 0.2**
Road	9.94 ± 1.1	9.81 ± 0.1	6.73 ± 0.1	3.68 ± 0.2	5.44 ± 0.5	**2.14 ± 0.2**	7.45 ± 0.4	3.52 ± 0.2	2.21 ± 0.2	2.27 ± 0.2
Average	17.05 ± 3.6	7.19 ± 2.4	5.41 ± 0.1	4.62 ± 0.5	4.88 ± 1.0	4.19 ± 0.3	9.12 ± 0.6	4.16 ± 0.2	3.46 ± 0.1	**2.42 ± 0.1**

TABLE VI

ABUNDANCE ESTIMATION RESULTS ($\times 10^{-2}$) BY DIFFERENT UNMIXING METHODS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE IN BOLD

Abundance	VCA	$L_{1/2}$-NMF	Dgs-NMF	EndNet	SNMF-Net	TANet	CNNAEU	MiSiCNet	SSAE	MSSS-Net
Tree	16.09 ± 3.1	16.16 ± 0.5	11.66 ± 0.2	8.85 ± 0.4	8.62 ± 0.3	8.05 ± 0.4	13.56 ± 0.3	6.70 ± 0.3	5.35 ± 0.6	**4.01 ± 0.4**
Water	6.06 ± 1.8	5.57 ± 0.0	**4.13 ± 0.0**	6.87 ± 0.3	9.02 ± 0.3	8.13 ± 0.3	9.66 ± 0.3	6.18 ± 0.3	5.42 ± 0.4	4.62 ± 0.4
Soil	15.00 ± 1.5	17.02 ± 0.4	11.13 ± 0.3	10.55 ± 0.2	7.11 ± 0.2	8.59 ± 0.4	10.61 ± 0.2	10.15 ± 0.4	6.30 ± 0.4	**4.69 ± 0.3**
Road	11.09 ± 1.5	6.73 ± 0.2	5.68 ± 0.1	11.20 ± 0.4	7.73 ± 0.2	**5.45 ± 0.3**	8.64 ± 0.2	8.29 ± 0.4	7.12 ± 0.3	6.45 ± 0.3
Average	12.05 ± 1.5	11.37 ± 0.2	8.15 ± 0.2	9.37 ± 0.5	8.12 ± 0.3	7.56 ± 0.3	10.62 ± 0.1	7.83 ± 0.3	6.05 ± 0.2	**4.94 ± 0.2**

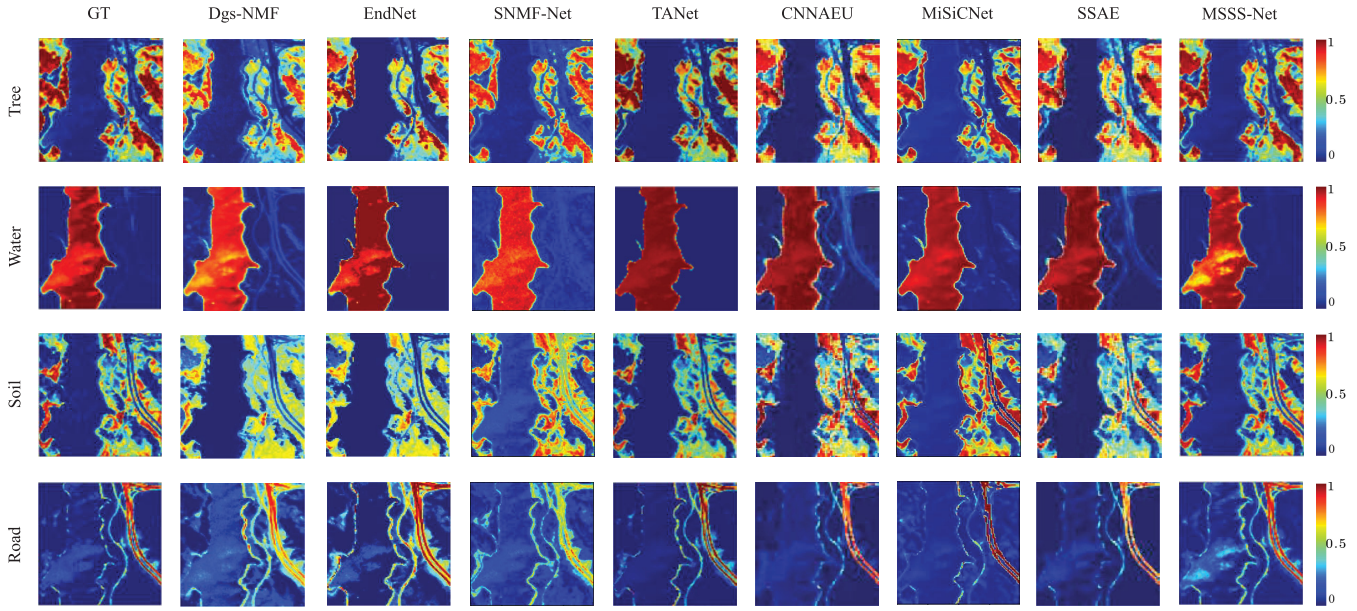


Fig. 10. Abundance maps estimated by different unmixing methods on the Jasper Ridge dataset and the corresponding GTs.

of the dataset and the 3-D convolution operation. A small dataset cannot fully exploit the power of 3-D convolution. The patch-based spatial information utilization strategy enables SSAE and MSSS-Net to extract higher quality endmembers. Furthermore, it is worth noting that the effect of applying spatial information on this dataset is much larger than that on the Urban dataset, which can be seen from the unmixing results of SSAE and MSSS-Net. We think that this may be related to the spatial resolution of the Jasper Ridge dataset. The lower spatial resolution of 20 m (2 m for Urban) results in a more heavily mixed degree of this dataset. In this case, the effect of applying spatial information is more obvious. The adoption of a recurrent perception strategy enables MSSS-Net to achieve greater improvements on this dataset.

Table VI shows the abundance estimation results of corresponding unmixing methods. We can see that CNNAEU and EndNet do not significantly improve over traditional methods. Unsatisfactory estimation results imply that unmixing with 3-D convolution still needs further development, while MiSiC-Net is an example that has achieved good results. Meanwhile, SSAE using 1-D spectral convolution achieves a considerable improvement in abundance. MSSS-Net with multiview cascaded RNNs' encoder achieves the best estimation result, which is around 33%–47% better than EndNet or TANet. This fully demonstrates the effectiveness of the multiview spectral information and recurrent fusion strategy for abundance estimation. In addition, the unmixing methods using spatial information have obtained an obvious improvement. This can be seen from the comparisons between $L_{1/2}$-NMF and Dgs-NMF, EndNet and SSAE, or MSSS-Net and SSAE, which shows a larger improvement compared to the results on the Urban dataset.

Fig. 10 visualizes all the abundance maps of the main comparison methods, which echoes the results in Table VI.

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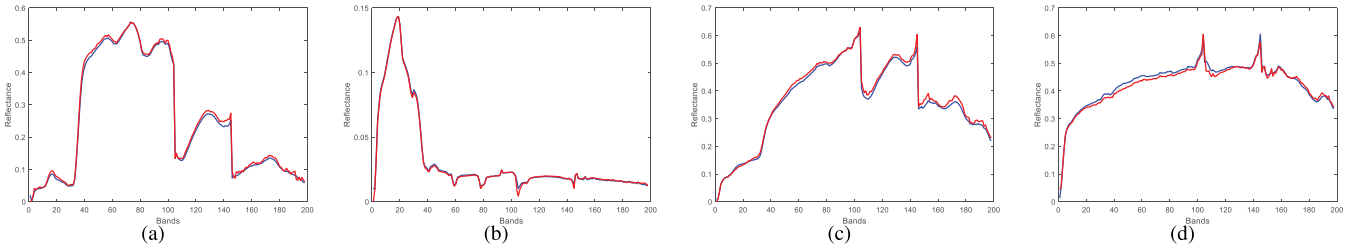


Fig. 11. Endmembers extracted by MSSS-Net (blue curves) on the Jasper Ridge dataset and the corresponding GTs (red curves). (a) Endmember Tree. (b) Endmember Water. (c) Endmember Soil. (d) Endmember Road.

TABLE VII

ENDMEMBER EXTRACTION RESULTS ($\times 10^{-2}$) BY DIFFERENT UNMIXING METHODS ON THE CUPRITE DATASET. THE BEST RESULTS ARE IN BOLD

Endmember (SAD)	VCA	$L_{1/2}$-NMF	Dgs-NMF	EndNet	SNMF-Net	TANet	CNNAEU	MiSiCNet	SSAE	MSSS-Net
Alunite	9.62 ± 1.5	14.64 ± 1.8	12.48 ± 1.8	14.56 ± 2.5	12.23 ± 2.5	12.98 ± 2.5	12.34 ± 5.0	11.45 ± 1.5	**9.15 ± 0.5**	12.12 ± 0.5
Andradite	10.56 ± 2.0	7.86 ± 1.4	7.59 ± 1.1	8.78 ± 0.5	9.34 ± 1.5	8.12 ± 0.3	10.89 ± 5.0	9.07 ± 0.5	9.34 ± 0.8	**7.35 ± 0.8**
Buddingtonite	11.71±1.5	11.84±5.6	**10.81±3.1**	12.45±1.2	13.45±2.0	10.89±1.0	12.45 ± 4.0	12.45 ± 1.0	14.89 ± 1.5	12.62 ± 1.0
Dumortierite	9.05 ± 1.5	11.53 ± 3.2	11.01 ± 2.1	7.89 ± 0.5	10.98 ± 1.6	9.12 ± 0.8	13.12 ± 3.0	**7.25 ± 0.6**	10.12 ± 1.0	8.20 ± 1.0
Kaolinite1	9.84 ± 3.0	9.68 ± 2.3	9.02 ± 2.8	11.01 ± 1.3	8.56 ± 1.8	11.89 ± 1.0	11.23 ± 5.0	10.12 ± 0.8	**7.34 ± 0.8**	10.81 ± 1.0
Kaolinite2	7.55 ± 1.6	7.24 ± 0.9	7.42 ± 1.2	13.45 ± 0.5	9.67 ± 2.5	6.12 ± 0.3	8.55 ± 3.5	12.37 ± 1.3	7.45 ± 0.5	**5.19 ± 0.5**
Muscovite	13.66 ± 3.6	25.12 ± 5.5	20.51 ± 6.0	9.12 ± 1.5	15.56 ± 3.0	14.78 ± 1.0	9.99 ± 3.5	**8.38 ± 0.5**	9.01 ± 0.5	11.57 ± 0.8
Montmorillonite	7.01 ± 1.0	7.46 ± 1.5	7.27 ± 1.2	8.67 ± 0.3	6.78 ± 1.0	6.02 ± 0.3	6.12 ± 3.6	7.97 ± 0.3	8.89 ± 0.6	**5.44 ± 0.5**
Nontronite	9.21 ± 2.5	9.54 ± 2.3	8.88 ± 1.7	11.23 ± 0.2	9.12 ± 2.5	8.78 ± 0.5	10.56 ± 3.0	10.31 ± 1.0	9.12 ± 0.8	**7.38 ± 1.0**
Pyrope	11.87 ± 5.0	8.94 ± 5.2	8.82 ± 5.0	7.45 ± 0.5	9.15 ± 1.3	**6.23 ± 0.3**	8.98 ± 4.0	6.85 ± 0.4	7.89 ± 1.0	7.44 ± 1.5
Sphene	19.30 ± 7.0	7.91 ± 2.5	8.15 ± 2.1	7.01 ± 1.5	7.99 ± 1.5	9.78 ± 1.5	7.12 ± 3.0	7.44 ± 1.0	7.12 ± 1.5	**6.86 ± 0.5**
Chalcedony	12.56 ± 1.0	13.76 ± 6.3	13.62 ± 6.3	12.12 ± 0.8	**6.12 ± 0.6**	9.34 ± 1.5	15.16 ± 4.0	10.15 ± 1.5	12.12 ± 2.0	14.73 ± 1.5
Average	10.99 ± 1.5	11.29 ± 1.0	10.46 ± 0.5	10.31 ± 0.8	9.91 ± 0.6	9.50 ± 1.0	10.54 ± 3.0	9.48 ± 1.0	9.37 ± 0.8	**9.14 ± 0.6**

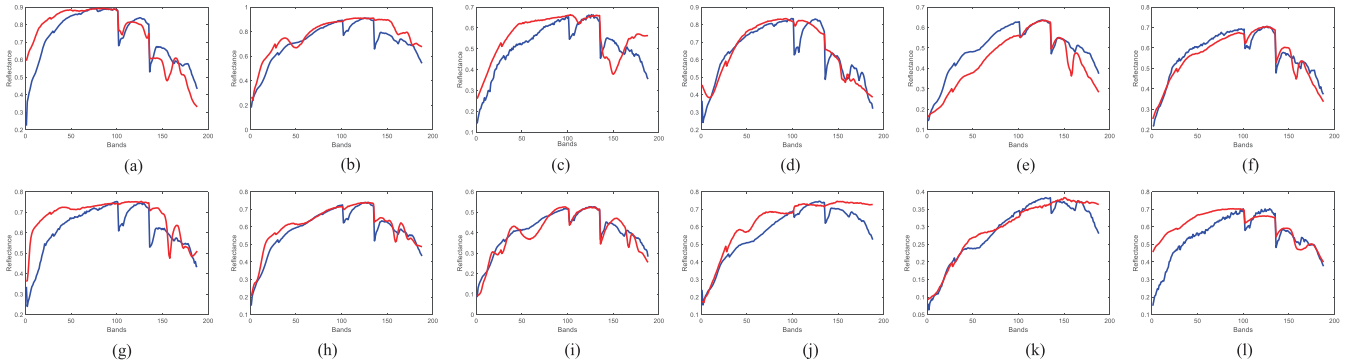


Fig. 12. Endmembers extracted by MSSS-Net (blue curves) on the Cuprite dataset and the corresponding GTs (red curves). (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite$_1$. (f) Kaolinite$_2$. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

We can see that the unmixing methods show more differences in the abundances of Soil and Road endmembers, where AE-based unmixing methods obtain better abundance maps overall. The reason is that these two endmembers are relatively similar, and their degree of mixing and sparseness is relatively high. The abundance maps of both SSAE and MSSS-Net are close to GT, while MSSS-Net exhibits more abundance details. On the other hand, Fig. 11 visualizes the endmember spectral curves of MSSS-Net and GT. It can be seen that the four endmember curves are very close to GTs, which shows the effectiveness of the patch-based spatial information and the cascaded RNNs' encoder.

4) Experiments on Cuprite Dataset: Table VII presents the endmember extraction results of all unmixing methods on the Cuprite dataset. Although the Cuprite dataset is a more complex HSI with 12 endmembers, from Table VII, we can conclude that all unmixing algorithms obtain satisfactory endmembers for several materials, such as Andradite, Kaolinite1, Montmorillonite, and Nontronite. Most importantly, our proposed MSSS-Net achieves the best average SAD value. In terms of individual endmembers, the MSSS-Net gets the best score on half of the materials, which fully illustrates the superiority of the proposed patch-based cascaded RNN encoder.

Fig. 12 visualizes the USGS reference signatures (GTs) along with endmember spectral curves extracted by MSSS-Net on the Cuprite dataset. Corresponding to the SAD values in Table VII, from Fig. 12, it can be seen that the estimated endmembers are generally well matched with the GTs. These results illustrate the effectiveness of MSSS-Net on complex hyperspectral data. On the other hand, since the GT for true abundances is not available, we only present the abundance estimation results of MSSS-Net, as shown in Fig. 13. It can be seen that there is no overlapping region, which indicates a satisfactory unmixing result in this particular scenario.

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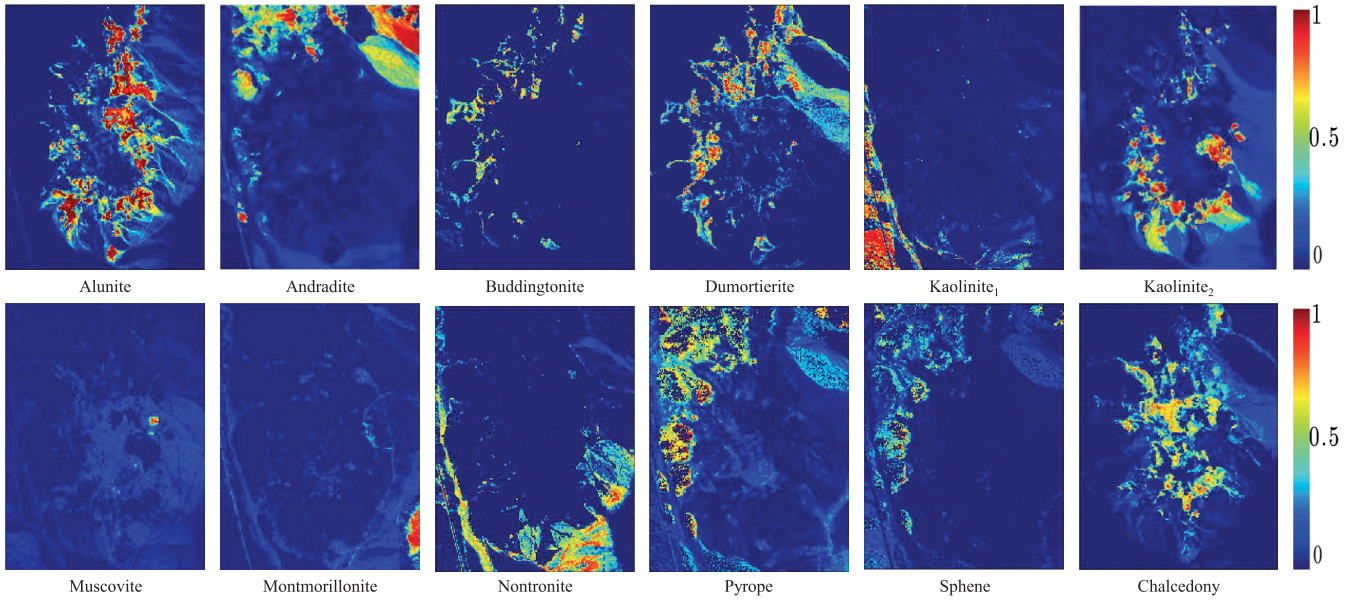


Fig. 13. Abundance maps estimated by MSSS-Net on the Cuprite dataset.

TABLE VIII

ENDMEMBER EXTRACTION AND ABUNDANCE ESTIMATION RESULTS ($\times 10^{-2}$) ON THE SYNTHETIC DATASET FOR THE SSR NETWORK. THE BEST RESULTS ARE IN BOLD

	Endmember (SAD)			Abundance (RMSE)		
Endmember	SSAE (EENet)	SSR w/o SW	SSR w/ SW	SSAE (EENet)	SSR w/o SW	SSR w/ SW
# 1	4.49 ± 0.2	3.76 ± 0.2	**3.65 ± 0.2**	13.83 ± 2.0	11.80 ± 1.8	12.15 ± 1.5
# 2	5.66 ± 0.3	4.08 ± 0.1	**4.06 ± 0.2**	9.51 ± 1.5	9.89 ± 1.5	9.55 ± 1.0
# 3	4.13 ± 0.2	2.84 ± 0.1	**2.68 ± 0.1**	16.36 ± 2.0	16.61 ± 2.0	15.43 ± 2.0
# 4	4.41 ± 0.1	3.58 ± 0.1	**3.45 ± 0.1**	13.25 ± 1.5	12.31 ± 1.0	12.15 ± 1.0
# 5	3.62 ± 0.1	3.25 ± 0.2	**3.21 ± 0.1**	18.31 ± 1.8	15.92 ± 1.0	16.37 ± 0.8
Average	4.46 ± 0.1	3.50 ± 0.1	**3.41 ± 0.1**	14.25 ± 1.0	13.31 ± 0.8	13.13 ± 0.8

D. Ablation Study

The proposed two-stream MSSS-Net consists of two branch networks, SSR and MSSR, both of which can act as independent unmixing networks and construct an efficient network by sharing a decoder. We discuss the improvements that benefit from the two key concepts in this article: the cascaded RNNs' encoder and multiview spatial–spectral information. Meanwhile, we make some clarifications on the weight-sharing strategy.

1) Experiments on SSR Network: In the SSR network, we propose a cascaded RNNs' encoder for the bidirectional recurrent perception of spatial information within a patch, which can extract higher quality endmember matrices. Table VIII gives the experimental results of the SSR network on the synthetic dataset with a 35-dB SNR, as well as the unmixing results of EENet in the SSAE network. It is worth noting that EENet and AENet, which constitutes the SSAE network, are two completely independent networks, which are used for endmember extraction and abundance estimation, respectively. The decoder weights of AENet are not trainable, which are fixed as the endmember matrix estimated by EENet. Table VIII illustrates the experimental results of the independent network EENet. We point out that the EENet and independent SSR network (SSR w/o SW) are both patch-based unmixing networks, and its difference lies in the encoder, where EENet adopts the fully connected layers and SSR adopts a cascade RNNs. As can be seen in Table VIII, SSR w/o SW improves both endmember and abundance estimations, which shows the effectiveness of the cascaded RNNs' encoder. Furthermore, the SSR network (SSR w/ SW) further improves the performance after introducing the shared weight strategy.

2) Experiments on MSSR Network: In the MSSR network, we introduce the multiview spectral information and use a cascaded RNNs' encoder to bidirectionally cyclically fuse the spectral information in a pixel for better abundance representations. Table IX shows the experimental results of MSSR and SSAE on the synthetic dataset with a 35-dB SNR. It also shows the unmixing results of the independent network AENet and the fixed weight AENet, namely, AENet FW. It is worth noting that AENet FW is the SSAE network, and the fixed weight strategy is also used in CNNAEU.

We can observe from Table IX that neither AENet nor MSSR w/o SW obtains satisfactory endmembers and abundances, when used as independent unmixing networks. However, MSSR w/o SW obtains significantly better endmembers and abundances than AENet. This indicates the effectiveness of the multiview spectral information and recurrent fusion strategy. Furthermore, when the weights are fixed as endmembers extracted by EENet, the abundances estimated by AENet FW are greatly improved. After introducing the shared

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TABLE IX

ENDMEMBER EXTRACTION AND ABUNDANCE ESTIMATION RESULTS ($\times 10^{-2}$) ON THE SYNTHETIC DATASET FOR THE MSSR NETWORK. THE BEST RESULTS ARE IN BOLD

	Endmember (SAD)				Abundance (RMSE)			
Endmember	SSAE (AENet)	SSAE (AENet FW)	MSSR w/o SW	MSSR w/ SW	SSAE (AENet)	SSAE (AENet FW)	MSSR w/o SW	MSSR w/ SW
# 1	13.64 ± 2.5	4.49 ± 0.2	6.46 ± 0.4	**3.65 ± 0.2**	17.46 ± 2.0	7.21 ± 0.4	7.77 ± 1.5	**6.05 ± 0.2**
# 2	8.03 ± 0.8	5.66 ± 0.3	6.61 ± 0.2	**4.06 ± 0.2**	7.29 ± 0.9	7.58 ± 0.2	8.62 ± 0.5	**6.54 ± 0.1**
# 3	9.96 ± 2.2	4.13 ± 0.2	6.88 ± 0.3	**2.68 ± 0.1**	16.89 ± 1.5	8.21 ± 0.3	11.68 ± 1.2	**6.67 ± 0.2**
# 4	13.58 ± 2.0	4.41 ± 0.1	6.58 ± 0.3	**3.45 ± 0.1**	13.96 ± 1.5	8.01 ± 0.2	11.10 ± 1.0	**6.95 ± 0.1**
# 5	23.24 ± 5.2	3.62 ± 0.1	6.27 ± 0.2	**3.21 ± 0.1**	13.68 ± 2.5	9.09 ± 0.2	15.93 ± 2.0	**8.39 ± 0.2**
Average	13.69 ± 2.5	4.46 ± 0.1	6.56 ± 0.3	**3.41 ± 0.1**	13.86 ± 1.2	8.02 ± 0.2	11.02 ± 1.0	**6.92 ± 0.1**

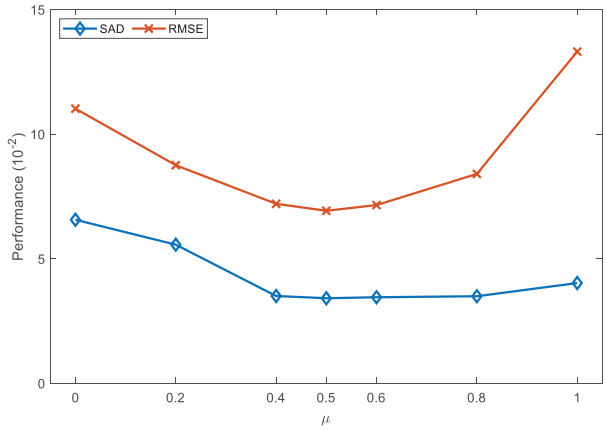


Fig. 14. Ablation experiments (mean SAD and RMSE) with respect to μ values.

weight strategy, MSSR w/ SW shares a decoder with the SSR network. Driven by higher quality endmember matrices, MSSR produces the best abundance estimation results.

3) Discussion: By comparing Tables VIII and IX, we can see that SSAE and MSSS-Net still generally follow the findings in [39]. In the spatial information-based AE network, both EENet and SSR networks can extract good endmembers, while their estimated abundances are unsatisfactory. On the other hand, in the AE network based on spectral information, AENet and MSSR networks still obtain better abundance results than EENet and SSR in the case of poor endmember extraction. After fixing the endmember matrix or adopting the shared weight strategy, the abundance estimation results will be further improved. In other words, the utilization of spatial information facilitates endmember extraction, while the usage of spectral information is more helpful for abundance estimation.

It is worth noting that, unlike in [39], we do not employ the abundance collaborative loss but only a weight-sharing strategy. We find that, in MSSS-Net, abundance synergy occasionally has opposite effects and results in poor endmember extraction. Meanwhile, Fig. 14 shows the ablation experiment on the value of μ, i.e., the performance of MSSS-Net with respect to μ. It can be seen that 0.4–0.6 is the most suitable interval. Furthermore, we have found that the weight-sharing strategy has little effect on the SSR network. After introducing the weight-sharing strategy, the improvements of both endmember and abundance results are limited. However, it has a strong impact on the MSSR network, and better endmembers have contributed to a large improvement in the abundance estimation, which is also reflected in Fig. 14. This strategy gives full play to the respective potentials of the two-steam network. As shown by the unmixing results, MSSS-Net achieves the best results on both endmembers and abundances. Meanwhile, we find that the estimated endmembers and abundances of the independent MSSR network, i.e., MSSR w/o SW, are still comparable. This again demonstrates the superiority of the proposed cascaded RNNs' encoder and multiview unmixing.

V. CONCLUSION

In this article, we explore multiview spatial–spectral information for HSI unmixing using an AE-based network, called MSSS-Net. It consists of a patch-based spatial RNN and a pixel-based multiview spectral RNN. Meanwhile, we design a cascaded RNNs' encoder to efficiently extract deep patch-pixel features in HSI, which provides a perspective on using only RNNs to build unmixing networks. The proposed two-stream network shares a decoder and is collaboratively trained in an end-to-end manner. Experiments and ablation studies on several hyperspectral data demonstrate the excellent performance of the proposed MSSS-Net. Despite the encouraging unmixing results obtained in this article, there are still some issues worthy of further investigation. The MSSS-Net adopts a general SAD-based loss function, and a more efficient reconstruction loss can be designed. In addition, for multiview unmixing, more datasets, especially ultraspectral images with a higher spectral resolution, will be used to further evaluate the unmixing performance of MSSS-Net.

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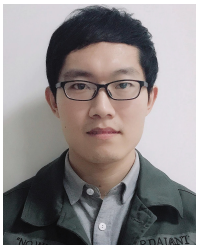
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